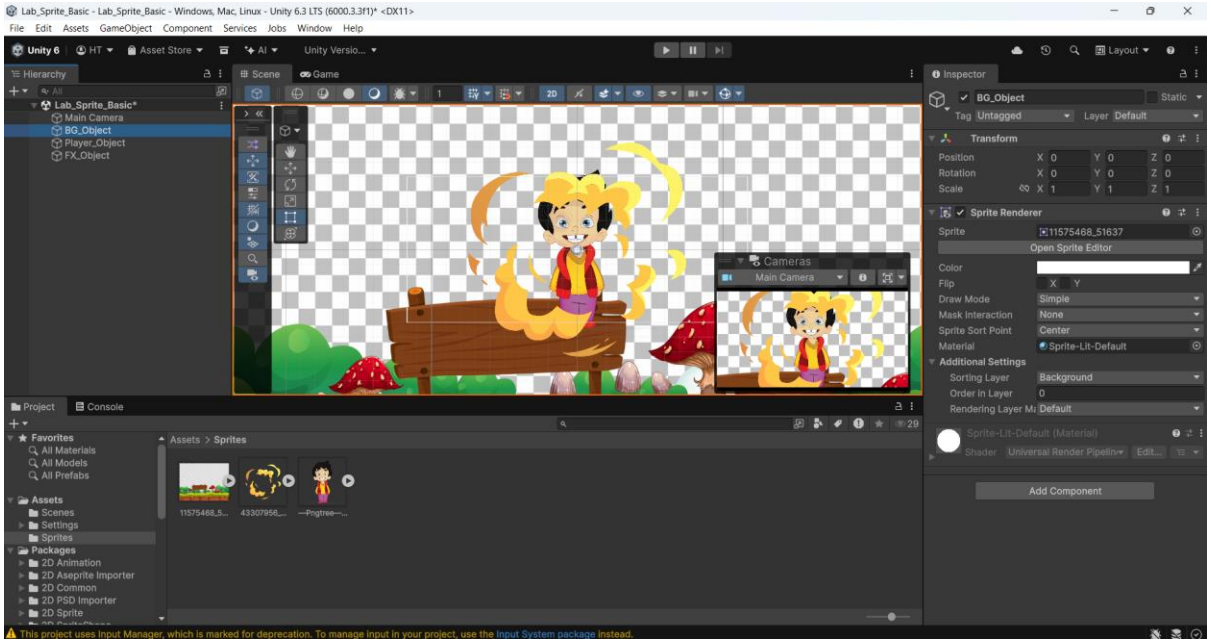
LAB THỰC HÀNH – CHƯƠNG 6: SPRITES & ANIMATIONS (UNITY)

Lab 1 – Sprite Creator & Sprite Renderer (Placeholder → Asset thật)

Mục tiêu: làm quen Sprite Creator và Sprite Renderer.
Yêu cầu:
- Tạo scene Lab_Sprite_Basic.
- Tạo 3 placeholder sprites bằng Sprite Creator (Square/Circle/Triangle).
- Tạo Sorting Layer: Background / Player / FX. Đặt Order in Layer khác nhau.
- Thay thế placeholder bằng sprite thật (drag sprite vào Sprite Renderer -> Sprite).

Deliverable: ảnh scene thể hiện Sorting Layer/Order hoạt động đúng.


Lab_Sprite_Basic - Lab_Sprite_Basic - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.3f1)* <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Hierarchy
Lab_Sprite_Basic*
Main Camera
BG_Object
Player_Object
FX_Object
Scene
Game
Cameras
Main Camera
Inspector
BG_Object
Static
Tag Untagged
Layer Default
Transform
Position X 0 Y 0 Z 0
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Sprite Renderer
Sprite 11575468_51637
Open Sprite Editor
Color
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Draw Mode Simple
Mask Interaction None
Sprite Sort Point Center
Material Sprite-Lit-Default
Additional Settings
Sorting Layer Background
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Sprite-Lit-Default (Material)
Shader Universal Render Pipelin
Edit...
Add Component
Project
Console
Favorites
All Materials
All Models
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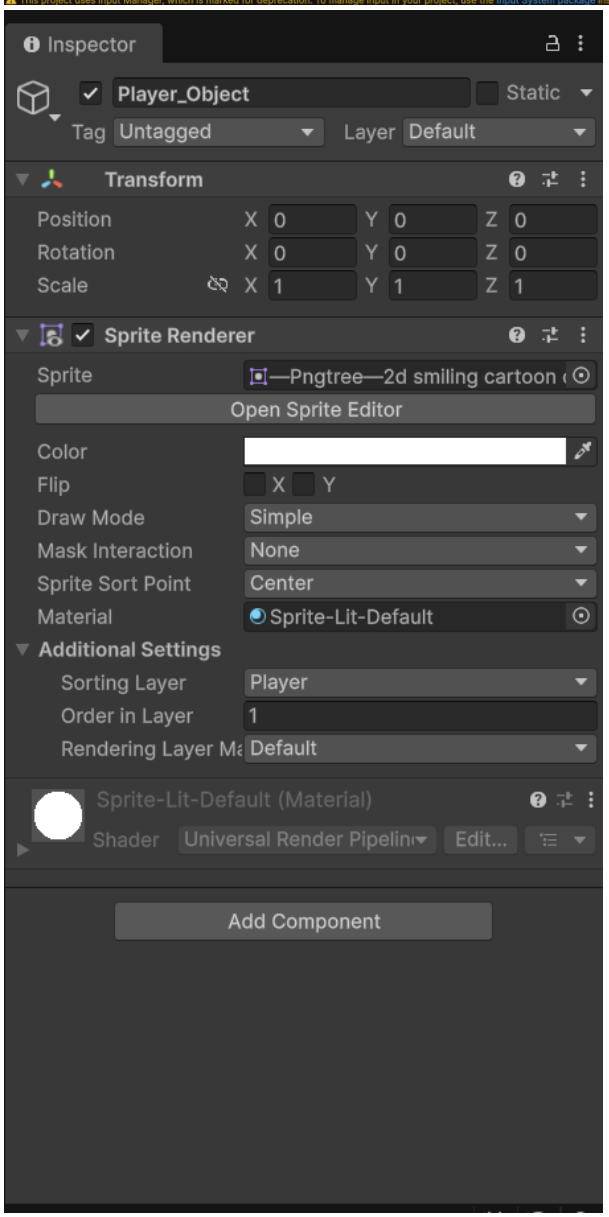
Assets
Scenes
Settings
Sprites
Packages
2D Animation
2D Aseprite Importer
2D Common
2D PSD Importer
2D Sprite
Assets > Sprites
This project uses Input Manager, which is marked for deprecation. To manage input in your project, use the Input System package instead.


Inspector
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Sprite Renderer
Sprite —Pngtree—2d smiling cartoon
Open Sprite Editor
Color
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Draw Mode Simple
Mask Interaction None
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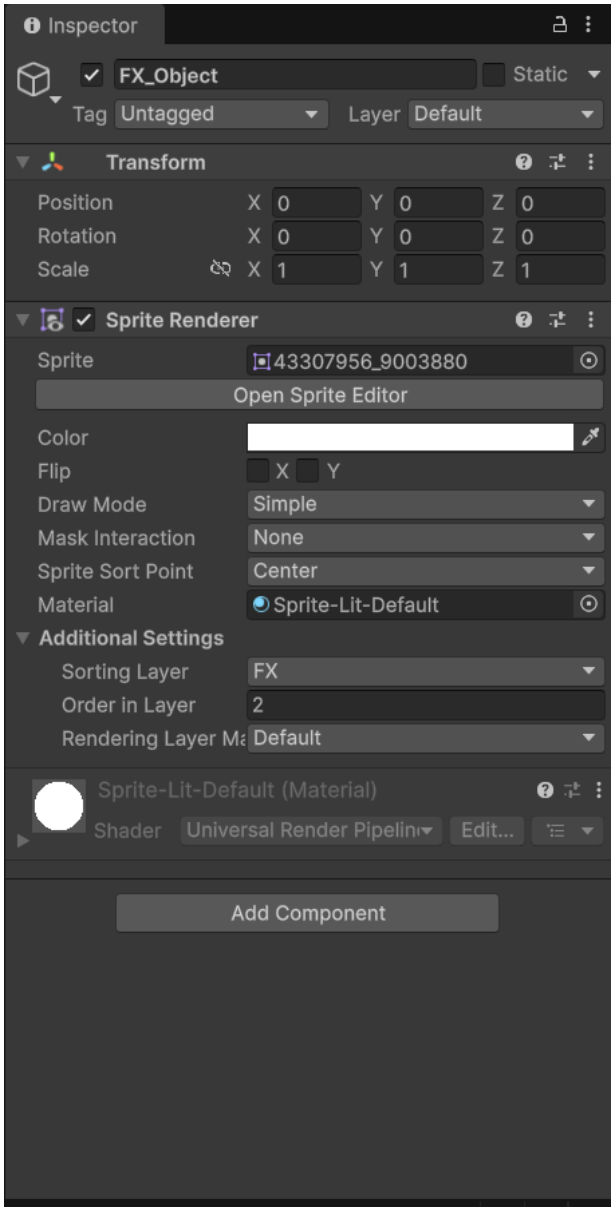
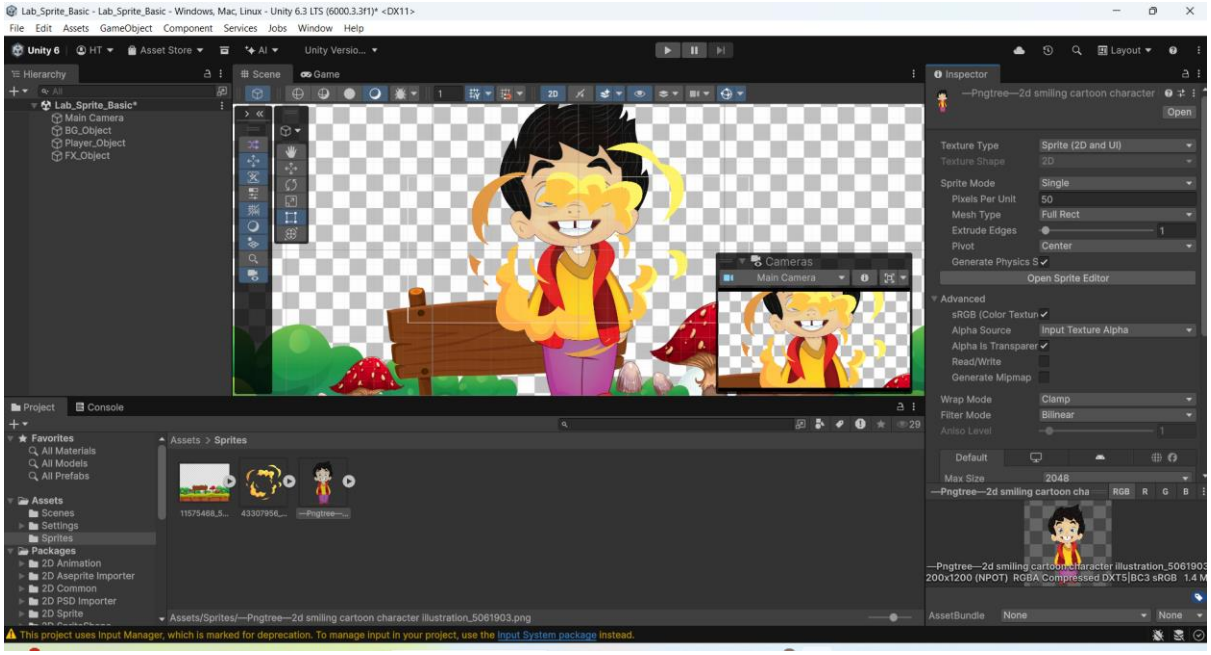


Lab 2 – Import Texture → Sprite (2D and UI) + Pixels Per Unit

Mục tiêu: cấu hình Import Settings cho sprite.
Yêu cầu:
- Import 2 ảnh PNG: 1 sprite đơn, 1 sprite sheet.
- Set Texture Type = Sprite (2D and UI).
- Thử 2 giá trị Pixels Per Unit khác nhau và quan sát kích thước trong Scene.
Deliverable: ảnh so sánh 2 PPU + giải thích ngắn PPU ảnh hưởng thế nào.


Lab_Sprite_Basic - Lab_Sprite_Basic - Windows, Mac, Linux - Unity 6.3 LTS (6000.3.3f1)* <DX11>
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Unity 6
Asset Store
Unity Versio...
Hierarchy
Scene
Game
Inspector
Lab_Sprite_Basic*
Main Camera
BG_Object
Player_Object
FX_Object
Cameras
Main Camera
—Pngtree—2d smiling cartoon character
Open
Texture Type
Sprite (2D and UI)
Texture Shape
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Sprite Mode
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Pixels Per Unit
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Mesh Type
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2D Aseprite Importer
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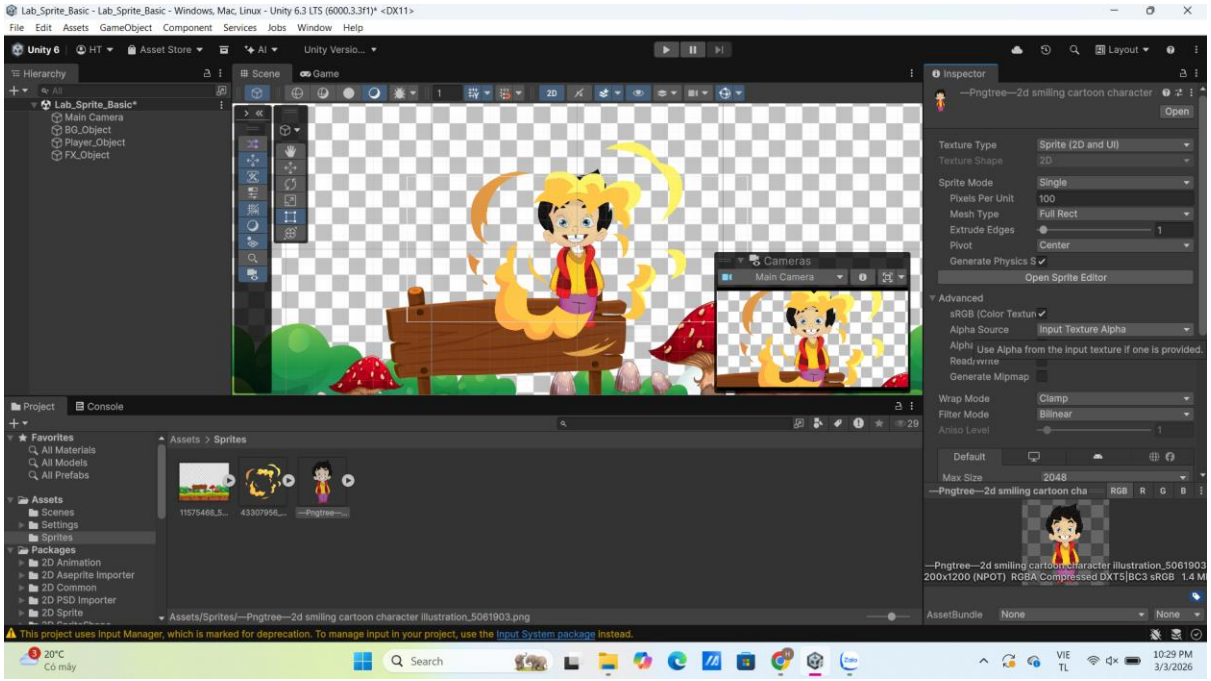
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Sprite (2D and UI)
Texture Shape
2D
Sprite Mode
Single
Pixels Per Unit
100
Mesh Type
Full Rect
Extrude Edges
Pivot
Center
Generate Physics S
Open Sprite Editor
Advanced
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Alpha Source
Input Texture Alpha
Alpha Is Transparer
Use Alpha from the input texture if one is provided.
Read/Write
Generate Mipmap
Wrap Mode
Clamp
Filter Mode
Bilinear
Aniso Level
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Max Size
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—Pngtree—2d smiling cartoon cha
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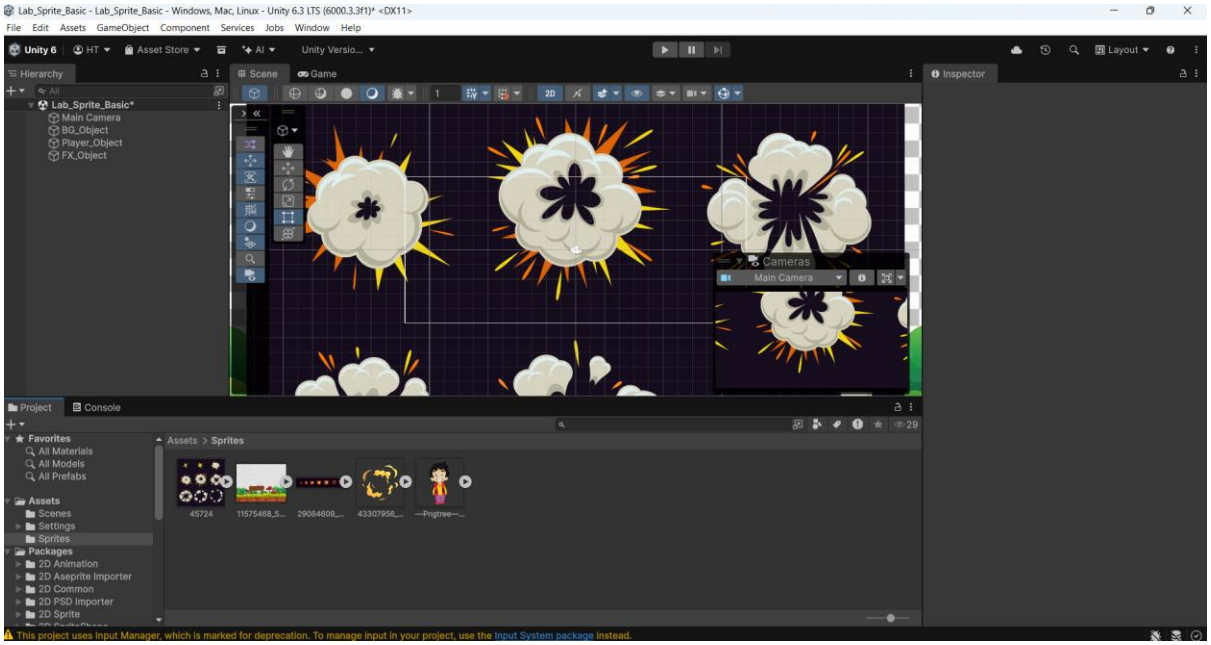
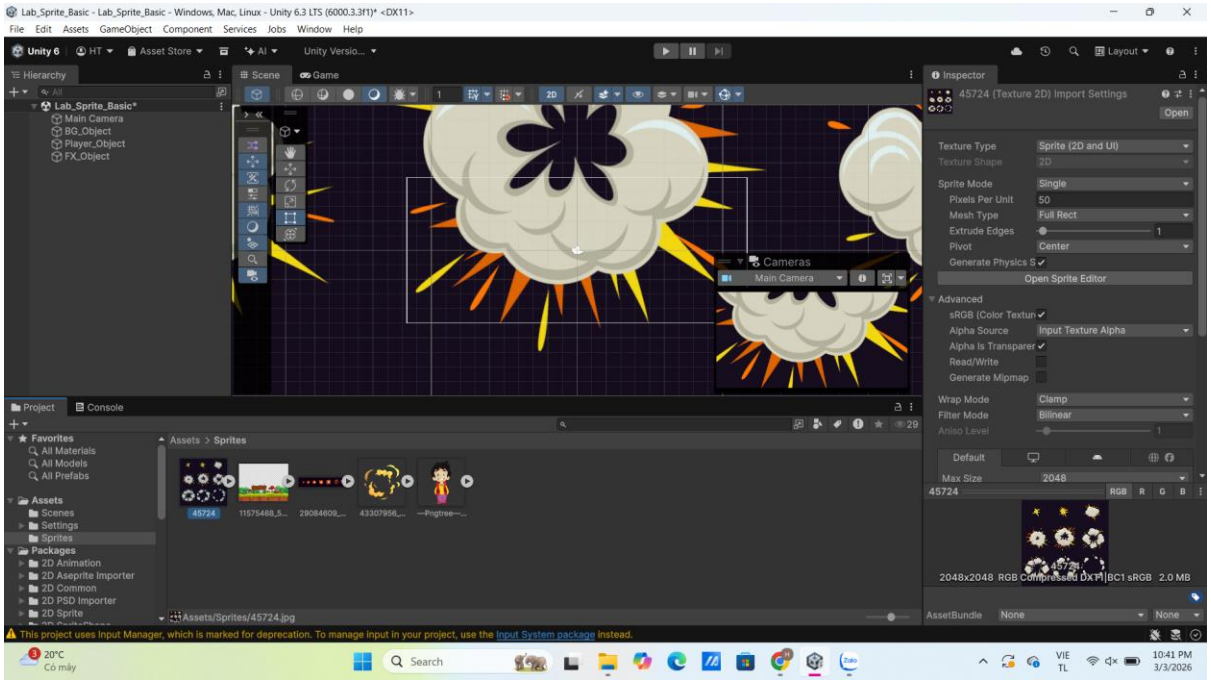
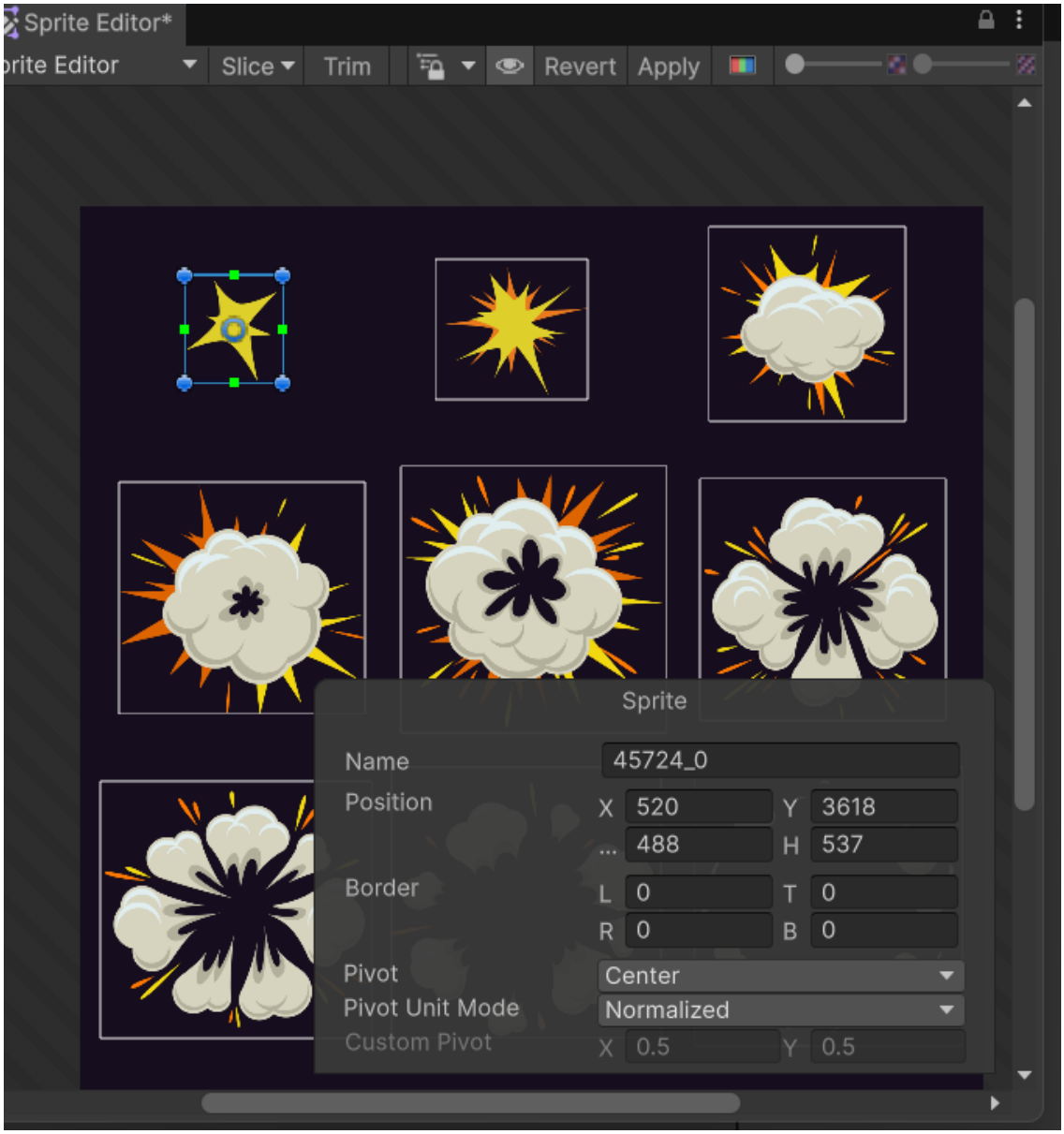




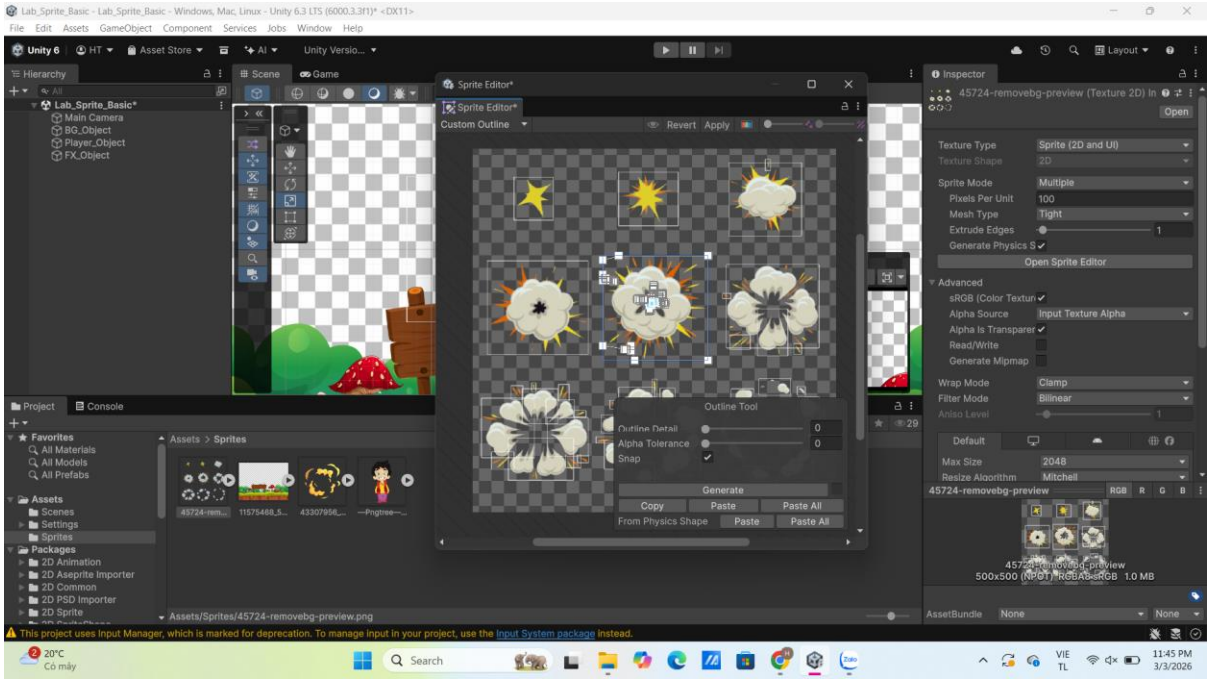
Lab 3 – Sprite Editor: Multiple Sprite Mode + Manual Slicing

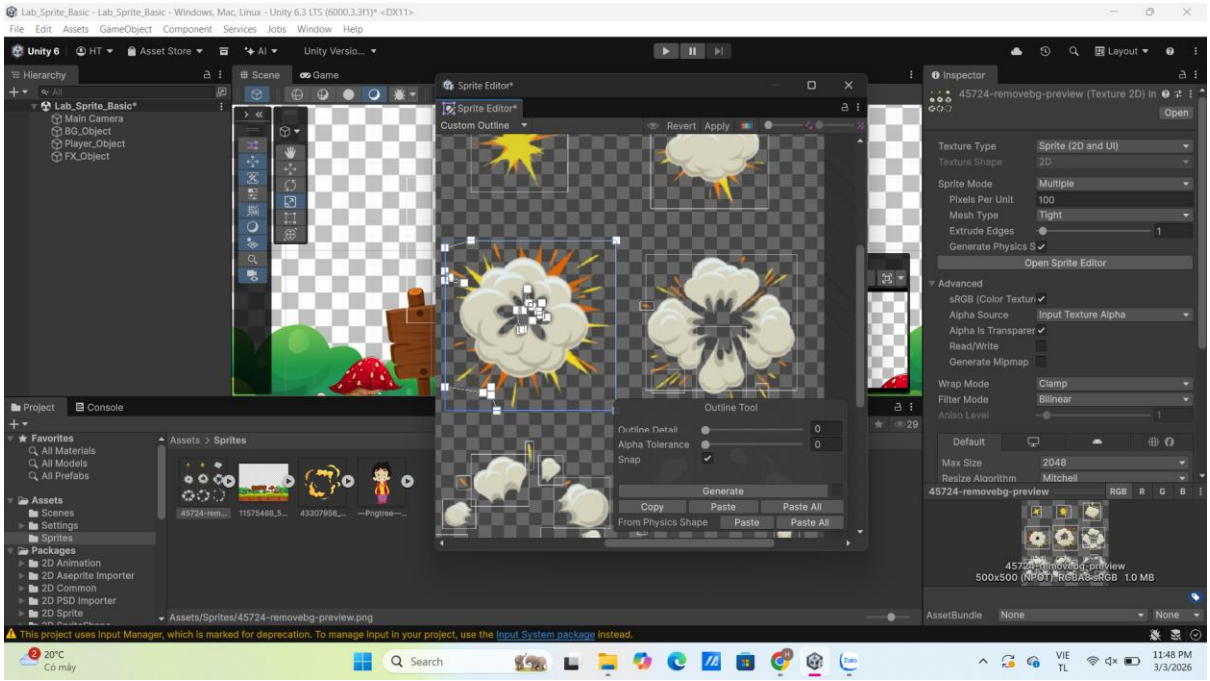
Mục tiêu: cắt sprite sheet thủ công và đặt pivot.
Yêu cầu:
- Set Sprite Mode = Multiple.
- Mở Sprite Editor, tạo các rectangle selection để tách ít nhất 6 sprite.
- Dùng Trim để khít theo transparency.
- Đặt Pivot: Center cho sprite thường, Bottom-Center cho sprite nhân vật.
Deliverable: ảnh Sprite Editor với các vùng slice + danh sách sprite đã cắt.



Lab 4 – Sprite Editor: Automatic/Grid Slicing + Edit Outline

Mục tiêu: dùng Auto/Grid slice và chỉnh outline để tối ưu mesh.
Yêu cầu:
- Dùng Automatic slicing (theo transparency) hoặc Grid by Cell Size/Count.
- Chọn 1 sprite và dùng Edit Outline để chỉnh outline (giảm điểm/đường bao).
Deliverable: ảnh before/after outline + nhận xét khi nào cần edit outline.





Automatic slicing tạo outline theo alpha nhưng thường sinh nhiều điểm dư.

Edit Outline giúp giảm số điểm polygon, tối ưu mesh và cải thiện hiệu suất.

Cần chỉnh outline khi sprite có hình phức tạp hoặc dùng trong physics/collider để tránh tính toán dư thừa.

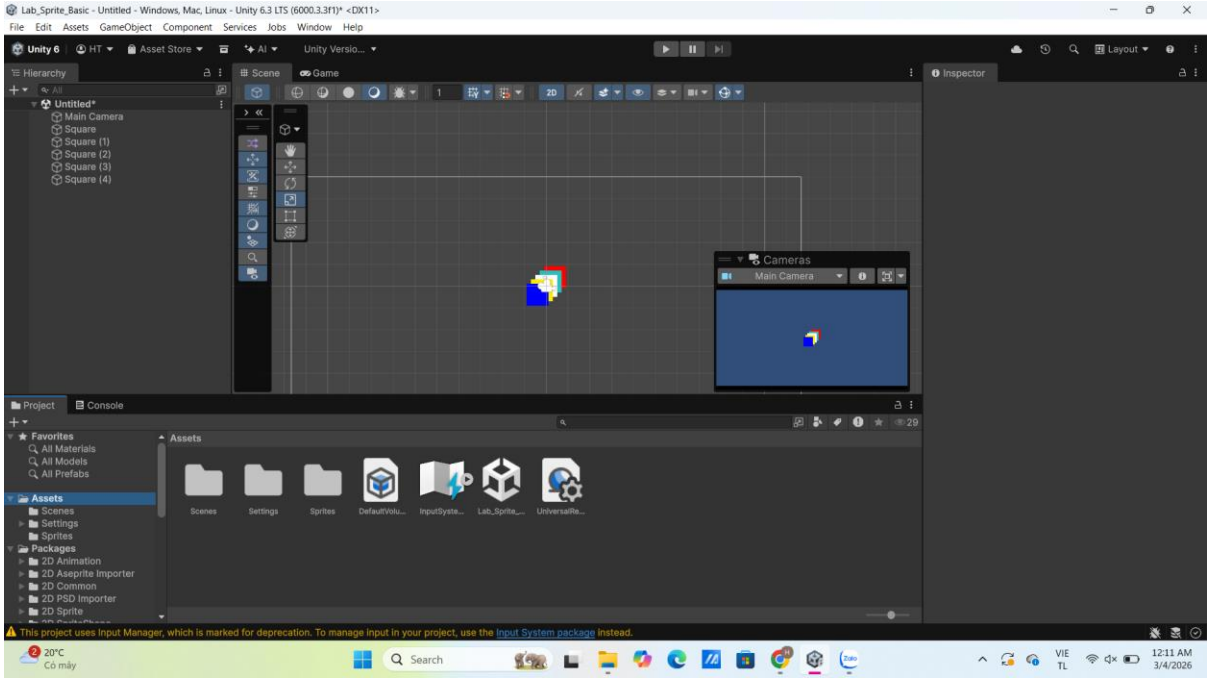
Lab 5 – Transparency Sort Mode (Orthographic/Perspective/Custom Axis)

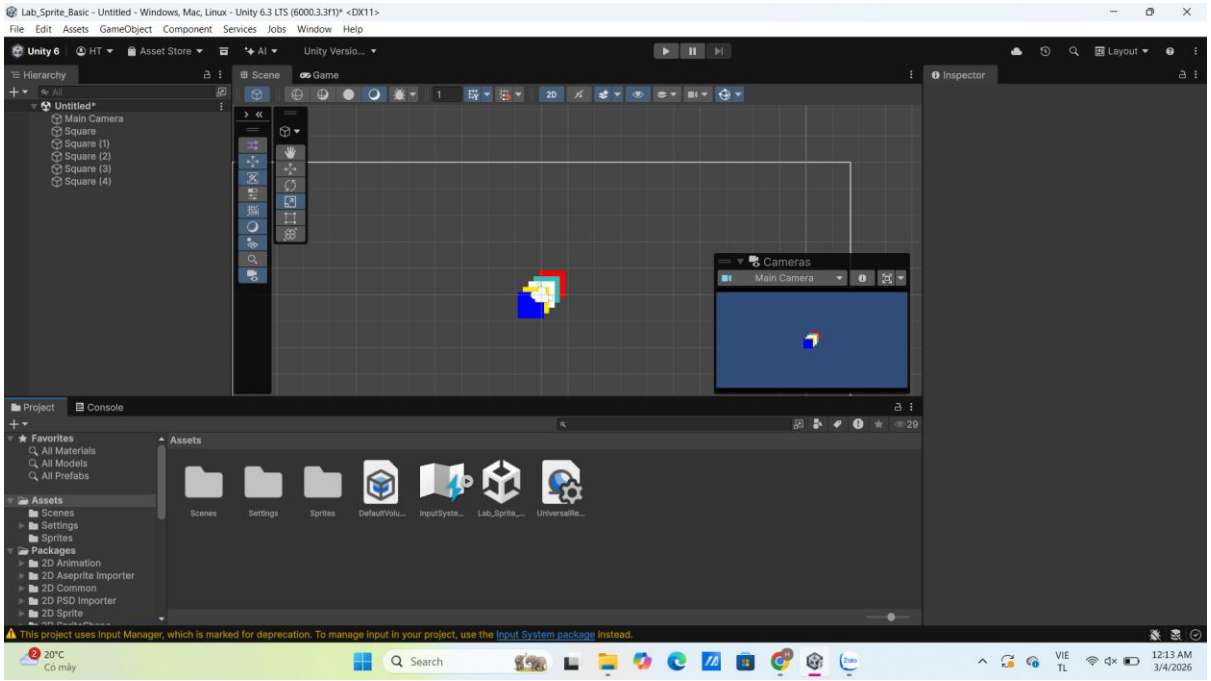
Mục tiêu: hiểu cách Unity sắp xếp sprite theo camera.

Yêu cầu:

- Tạo 1 scene có 5 sprite đặt lệch nhau theo trục Z (hoặc theo custom axis).
- Thử các mode: Default, Orthographic, Perspective, Custom Axis.
- Ghi lại kết quả (sprite nào nằm trên).

Deliverable: 2 ảnh minh họa khác biệt giữa Orthographic và Custom Axis.





Orthographic mode sắp xếp sprite theo trục Z.

Custom Axis cho phép sắp xếp theo trục tùy chỉnh (ví dụ Y), phù hợp với game 2D top-down.

Custom Axis giúp tạo hiệu ứng chiều sâu mà không cần thay đổi Z.